

함성함수

유형 01) 대응 함수

$$\textcircled{1} f(1)=3, f(2)=1 \\ g(1)=5, g(3)=2$$

$$\textcircled{1} (g \circ f)(1) = g(f(1)) = g(3) = 2$$

$$\textcircled{2} (g \circ f)(2) = g(f(2)) = g(1) = 5$$

$$\textcircled{2} f(x) = x^2 - 1, g(x) = 2x + 3$$

$$\textcircled{1} (g \circ f)(2) = g(f(2))$$

$$= g(4-1) = 9$$

$$\textcircled{2} (f \circ g)(2) = f(g(2))$$

$$= f(2 \cdot 2 + 3) = f(7)$$

$$= 49 - 1 = 48$$

$$\textcircled{3} f(x) = 3x - 2, g(x) = x^2$$

$$\textcircled{1} (g \circ f)(1) = g(f(1))$$

$$= g(1) = 1$$

$$\textcircled{2} (f \circ g)(1) = f(g(1))$$

$$= f(1) = 1$$

$$\textcircled{4} f(1)=2, f(2)=3, f(3)=1$$

$$\textcircled{1} (f \circ f)(1) = f(f(1)) = f(2) = 3$$

$$\textcircled{2} (f \circ f \circ f)(1) = f(f(f(1))) = f(f(2)) = f(3) = 1$$

$$\textcircled{5} f(x) = x + 4, g(x) = 2x$$

$$\textcircled{1} (g \circ f)(3) = g(f(3)) = g(7) = 14$$

$$\textcircled{2} (f \circ g)(3) = f(g(3)) = f(6) = 10$$

$$\rightarrow 14, 10$$

유형 02) 식으로 주어진 함수

$$\textcircled{6} f(x) = 2x + 1, g(x) = x^2 - x$$

$$\textcircled{1} (g \circ f)(2) = g(f(2)) = g(5) = 20$$

$$\textcircled{2} (f \circ g)(2) = f(g(2)) = f(2) = 5$$

$$\textcircled{7} f(x) = -2x^2 + 5x + 1$$

$$g(x) = \sqrt{2x-1}$$

$$\textcircled{1} (g \circ f)(1) = g(f(1)) = g(-2+5+1)$$

$$= g(4) = \sqrt{8-1} = \sqrt{7}$$

$$\textcircled{2} (f \circ g)(1) = f(g(1)) = f(\sqrt{1})$$

$$= -2+5+1 = 4$$

$$\textcircled{3} (g \circ g)(1) = g(g(1)) = g(1)$$

$$= 1$$

$$\textcircled{8} f(x) = -x^2 + 3x, g(x) = \sqrt{x+1}$$

$$\textcircled{1} (g \circ f)(1) = g(f(1)) = g(2) = \sqrt{3}$$

$$\textcircled{2} (f \circ g)(1) = f(g(1)) = f(\sqrt{2})$$

$$= -(\sqrt{2})^2 + 3\sqrt{2} = -2 + 3\sqrt{2}$$

$$\textcircled{9} f(x) = \frac{1}{x}, g(x) = x + 2 \quad (x > 0)$$

$$\textcircled{1} (g \circ f)(2) = g(f(2)) = g(\frac{1}{2}) = \frac{5}{2}$$

$$\textcircled{2} (f \circ g)(2) = f(g(2)) = f(4) = \frac{1}{4}$$

$$\textcircled{10} f(x) = |x-3|, g(x) = 2x-1$$

$$\textcircled{1} (f \circ g)(0) = f(g(0)) = f(-1) = 4$$

$$\textcircled{2} (g \circ f)(0) = g(f(0)) = g(3) = 5$$

유형 03) 합성함수 식구하기

o 11) $f(x) = x+3$, $g(x) = 2x-1$

① $(g \circ f)(x) = g(f(x)) = 2(x+3)-1$
 $= 2x+5$

② $(f \circ g)(x) = f(g(x)) = (2x-1)+3$
 $= 2x+2$

o 12) $f(x) = x^2$, $g(x) = x-4$

① $(g \circ f)(x) = g(f(x)) = x^2-4$

② $(f \circ g)(x) = f(g(x)) = (x-4)^2$

x 13) $f(x) = 4x^2-3$, $g(x) = \frac{1}{2} \cdot \sqrt{x+4}$

$(f \circ g)(x) = f(g(x)) = 4\left(\frac{1}{2} \cdot \sqrt{x+4}\right)^2 - 3$

$= 4\left(\frac{1}{2} \cdot (x+4)\right) - 3 = 2x + 8 - 3 = 2x+5$

$= 2(x+4) - 3 = 2x+8-3 = 2x+5$

o 14) $f(x) = 3x+1$, $g(x) = x^2-2x$

① $(g \circ f)(x) = g(f(x)) = (3x+1)^2 - 2(3x+1)$

$= 9x^2 + 6x + 1 - 6x - 2 = 9x^2 - 1$

② $(f \circ g)(x) = f(g(x)) = 3(x^2-2x) + 1$

$= 3x^2 - 6x + 1$

o 15) $f(x) = \frac{1}{x}$, $g(x) = x+2$ ($x \neq 0$, $x \neq -2$)

① $(g \circ f)(x) = g(f(x)) = \frac{1}{x} + 2$

② $(f \circ g)(x) = f(g(x)) = \frac{1}{x+2}$

o 13) $f(x) = 4x^2-3$, $g(x) = \frac{1}{2} \cdot \sqrt{x+4}$

$(f \circ g)(x) = f(g(x)) = 4\left(\frac{\sqrt{x+4}}{2}\right)^2 - 3$

$= 4x \cdot \frac{x+4}{4} - 3 = x+1$

유형 04) 자기 자신과 합성

x 16) $f(x) = -3x^2+2$, $f^2 = f \circ f$

$f(f(x)) = -3(-3x^2+2)^2 + 2$

$= -3(9x^4+4) + 2 = -27x^4 - 12 + 2$

$= -27x^4 - 10$

o 17) $f(x) = 2x$

① $f^3 = f \circ f \circ f = f(f(x)) = 2(2x) = 4x$

$f(2x) = 8x$

② $f^n(x) = 2^n x$

o 18) $f(x) = x+3$, $f^5(1)$

$f(x) = x+3$

$f^2 = (x+3)+3 = x+6$

$f^3 = (x+3)+6 = x+9$

$f^4 = (x+3)+9 = x+12$

$f^5 = (x+3)+12 = x+15$

$f^5(1) = 16$

x 19) $f(x) = x+k$, $f^3(2) = 1$

$f^1 = x+k$

$f^2 = (x+k)+k = x+2k$

$f^3 = (x+k)+2k = x+3k$

$2+3k = 1$, $3k = -1$, $k = -\frac{1}{3}$

o 20) $f(x) = x+k$, $f^4(3) = 15$

$f^1 = x+k$

$f^4 = x+4k$

$3+4k = 15$, $4k = 12$, $k = 3$

o 16) $f(x) = -3x^2+2$

$f \circ f = -3(-3x^2+2)^2 + 2$

$= -3(9x^4-12x^2+4) + 2$

$= -27x^4 + 36x^2 - 12 + 2$

$= -27x^4 + 36x^2 - 10$

o 19) $f(x) = x+k$, $f^3(2) = 11$

$f^1 = x+k$, $f^2 = x+2k$, $f^3 = x+3k$

$2+3k = 11$, $k = 3$